

	$-g = (10.0 \pm 1.0) \text{ m s}^{-2}$	
2(a)(i)	Magnitude of change in GPE $= (2.6)(0.0060) = 0.0156 \text{ J}$	[1] for correct substitution and correct answer
2(a)(ii)	Magnitude of change in EPE $= \frac{1}{2}(0.0060)(2.6 + 4.0) = 0.0198 \text{ J}$ <u>Common mistake:</u> <i>Calculate the change in EPE using $\frac{1}{2}k(0.6^2)$ instead of $\frac{1}{2}k(1.7^2 - 1.1^2)$.</i>	[1] for correct substitution [1] for correct answer
2(b)(i)	Total energy $= 0.0198 - 0.0156 = 0.0042 \text{ J}$	[1] for correct substitution
2(b)(ii)	$0.0042 = \frac{1}{2} \left(\frac{2.6}{9.81} \right) \omega^2 (0.60 \times 10^{-2})^2$ $\omega = 29.671 \text{ rad s}^{-1}$ $\omega \approx 30 \text{ rad s}^{-1}$	[1] for correct substitution [1] for correct answer
2(c)	Loss in total energy = Work done against air resistance $\frac{1}{2} \left(\frac{2.6}{9.81} \right) (29.671)^2 (0.0060^2 - 0.0040^2) = F_v (0.0060 + 0.0040)$ $F_v = 0.2333 \text{ N} = 0.23 \text{ N}$	[1] for correct application of conservation of energy [1] for correct substitution [1] for correct answer
3(a)	A longitudinal wave is a wave in which the oscillation of the particles	[1] for